Lactobacillus reuteri DSM 17938 in Infantile Colic: A Randomized, Double-Blind, Placebo-Controlled Trial

abstract

OBJECTIVE: To test the efficacy of *Lactobacillus reuteri* on infantile colic and to evaluate its relationship to the gut microbiota.

STUDY DESIGN: Fifty exclusively breastfed colicky infants, diagnosed according to modified Wessel's criteria, were randomly assigned to receive either *L reuteri* DSM 17 938 (10^8 colony-forming units) or pla- cebo daily for 21 days. Parental questionnaires monitored daily crying time and adverse effects. Stool samples were collected for microbiologic analysis.

RESULTS: Forty-six infants (*L reuteri* group: 25; placebo group: 21) completed the trial. Daily crying times in minutes/day (median [inter- quartile range]) were 370 (120) vs 300 (150) ($P = .127$) on day 0 and 35.0 (85) vs 90.0 (148) ($P = .022$) on day 21, in the *L reuteri* and placebo groups, respectively. Responders (50% reduction in crying time from baseline) were significantly higher in the *L reuteri* group versus pla- cebo group on days 7 (20 vs 8; $P = .006$), 14 (24 vs 13; $P = .007$), and 21 (24 vs 15; $P = .036$). During the study, there was a significant increase in fecal lactobacilli ($P = .002$) and a reduction in fecal *Escherichia coli* and ammonia in the *L reuteri* group only ($P = .001$). There were no differences in weight gain, stooling frequency, or incidence of consti- pation or regurgitation between groups, and no adverse events related to the supplementation were observed.

CONCLUSION: *L. reuteri* DSM 17 938 at a dose of 10^8 colony-forming units per day in early breastfed infants improved symptoms of infantile colic and was well tolerated and safe. Gut microbiota changes induced by the probiotic could be involved in the observed clinical improve- ment. *Pediatrics* 2010;126:e526–e533

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KEY WORDS

Lactobacillus reuteri, infantile colic, FISH, gut microflora, *Escherichia coli*

ABBREVIATIONS

FISH—fluorescence in situ hybridization
IQR—interquartile range

This trial has been registered at www.clinicaltrials.gov (identifier 00893711).

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